

STRENGTHS

- Huge demand forecast for all types of lighting products, specially LEDs
- Large distribution network with a large number of retail outlets
- Availability of vast pool of manpower (skilled/unskilled)
- Availability of natural resources, except rare earth materials
- Strong manufacturing capabilities and capacities for luminaires and light sources (GLS, CFL and FTL), which are ready for conversion to LED manufacturing plants

WEAKNESSES

- Low capability for domestic production of electronic components
- Inadequate government support for manufacturing compared to global competitors like China, which has become a global manufacturing hub for LED/electronic components through excellent government initiatives
- Low capabilities in R&D and testing labs
- Lack of skilled manpower
- Unavailability of rare earth materials
- High cost of capital due to high interest rates
- Low consumer confidence due to poor-quality products in the market
- Low awareness among price-sensitive consumers about the benefits of LEDs
- Poor power quality requires products to have very strong specifications
- Weak logistics infrastructure
- Inefficient and high transportation costs

SWOT

OPPORTUNITIES

- Huge potential to move LEDs and electronic components manufacturing to India
- India likely to become cost competitive in relation to China for the export of manufactured goods, due to increasing labour costs in China and a strengthening Yuan, coupled with a weakening rupee
- Potential to move to greater automation in lighting via a strict and mandatory energy conservation building code (ECBC); this would also leverage India's software skills for automation
- Fast-growing Indian infrastructure (projected CAGR of five per cent to seven per cent over the next five years)
- Scope for more environment-friendly products that can also be disposed of in a green way
- LED demand generation supported by various government-led procurement initiatives and developmental projects like JNN Solar Mission, DeitY's Electronics Policy, rural electrification programmes, 100 smart cities, UJALA and so on

THREATS

- Low-quality, cheap imports (for example, from China) flooding the market and competing with locally-manufactured products
- India faces an acute shortage of experts in fields like chemical, optics, lighting and thermal management. This expertise is primarily required for the manufacture of LED chips, which again calls for huge investments
- Disruptive technological changes, which require time and money for adoption by local manufacturing plants
- Adoption of protectionist policies by many countries, closing out the option of exporting locally-manufactured products
- Forex volatility, leading to fluctuating raw material costs for LEDs, most of which are imported today

Fig. 4: SWOT analysis of Indian LED industry

exponentially in India and the size of the market has also grown tremendously. Key drivers in the LED residential sector are energy efficiency, return on investment (RoI), low maintenance costs and environment friendliness. Energy Efficiency Services Ltd's (EESL's) remarkable initiatives to provide free LED lamps in rural areas and subsidised lamps in urban areas are ensuring large-scale adoption of this technology across the country.

Another segment that is expected to grow significantly in the coming years is outdoor lighting, as it would be backed by the government. The Centre has taken steps and is com-

mitted to installing LEDs for domestic lighting and streetlighting in 100 cities. The mandate to change 27.5 million streetlights to LEDs over the next few years will help in saving more than 5000MW of electricity consumption over a period of three years. Many states have already started to replace old streetlights.

This upward trend will continue and the projected demand is likely to increase mainly in outdoor (street and road) lighting. According to the Market and Research report, the smart streetlighting market in India is expected to grow at a CAGR of 42.2 per cent during the forecast period of 2016-2022.

In some parts of India, LED street-lighting projects have been implemented. But this is still at a nascent stage; it is expected that in the next five years, most parts of the country would have switched to smart streetlighting. Smart streetlighting poles automatically adjust the brightness according to the time of the day. These solutions can be operated manually as well as remotely.

For commercial projects like office-space and industrial-space lighting, RoI and energy efficiency become the most important aspects of the purchase decision. LEDs have the fastest RoI and help reduce energy consumption and light wastage. That is why